

Attempting To Restore Audio using WaveNet

Mihir Ranjan

May 28, 2026

1 Introduction

In the recent years, the major non-linear editing software such as Adobe Premiere Pro and Blackmagic's DaVinci Resolve Studio 20 have released "AI Audio Isolation" where the background noise is removed there respective AI Models, giving dialogue that is supposed to be smooth and "cinematic". However, each audio file varies due to the noise in the audio and how audio is recorded, resulting in hollow, robotic, or both sounding audio. In this repository, I attempt to use to restore noisy audio using WaveNet, a neural network architecture intended for training signals on the DAPS dataset.

2 Background

WaveNet is a deep generative model that processes audio on a single sample level, which relies on probability and autoregression. In terms of audio restoration, WaveNet acts as a probabilistic filter that predicts degraded audio samples by evaluating the past and future context. Audio is modeled sequentially as a joint probability distribution, which means that the probability of a specific audio sample x_t depends on

$$P(x) = \prod_{t=1}^T P(x_t \mid x_1, x_2, \dots, x_{t-1})$$

Because raw audio is usually 16-bit, WaveNet compresses the signal into discrete 8-bit integers using μ -law quantization

$$f(x_t) = \text{sgn} \frac{\ln(1 + \mu|x_t|)}{\ln(1 + \mu)}$$

where x_t is continuous audio samples normalized between -1 and 1 and $\mu = 255$

DAPS is a dataset of professionally produced studio speech recordings and recordings of the same speech on common consumer devices in real-world environments. This dataset contains 15 versions of audio with 3 professional versions and 12 consumer device.

3 Methodology

This repository uses the DAPS dataset as the paired clean and degraded speech dataset. Stated in the background, DAPS contains aligned professionally produced studio speech recordings and recordings of the same speech captured on consumer devices in real-world environments [2]. In this repository, the clean target condition was chosen as `cleanraw`, while the degraded input condition was chosen as `ipad.balcony1`. The audio files were then paired by matching the base file stem between the clean and degraded folders.

After filtering the files, there were 100 paired clean and degraded audio recordings. These pairs were split into 70 training pairs, 15 validation pairs, and 15 test pairs. All the audio was loaded as mono and resampled to a target sample rate of 16 kHz. Since the clean and degraded files are aligned versions of the same speech, each pair was trimmed to the same length before being used for training.

Instead of training on full audio recordings, the model was trained on short waveform chunks. Each chunk was one second long, corresponding to 16000 samples at the 16 kHz sampling rate. During training, random one-second chunks were sampled from each audio pair and validation and test examples used fixed chunks for consistency. The model therefore learned a supervised mapping

$$\text{degraded waveform} \longrightarrow \text{clean waveform}.$$

The model was a WaveNet-style restoration network inspired by the original WaveNet architecture of van den Oord et al. [1]. However, instead of using WaveNet as a fully autoregressive generator, the repository adapted it as a supervised audio restoration model. The model takes a degraded waveform chunk as input and directly outputs a restored waveform chunk of the same length.

The architecture used dilated one-dimensional convolutions, gated activations, residual connections, and skip connections. Dilated convolutions allow the model to capture longer-range temporal structure without requiring extremely large convolutional filters. The dilation pattern repeated across several residual blocks, allowing the network to observe both local waveform details and broader speech structure. The final output layer used a tanh activation so that the restored waveform remained in the normalized audio range.

The model used these hyperparameters:

- Input channels: 1
- Residual channels: 64
- Skip channels: 128
- Number of residual blocks: 3
- Number of dilation layers per block: 8
- Kernel size: 3
- Trainable parameters: 909,185
- Optimizer: Adam
- Learning rate: 10^{-4}
- Number of epochs: 50
- Batch size: 32

The training objective combined waveform level accuracy with frequency domain accuracy. The waveform loss (time domain) was the mean absolute error between the restored waveform and the clean target waveform. The spectral loss (frequency domains) was computed by comparing the log-magnitude short-time Fourier transforms of the restored and clean signals. The total loss was

$$\mathcal{L} = \mathcal{L}_{\text{waveform}} + 0.5\mathcal{L}_{\text{spectral}},$$

where

$$\mathcal{L}_{\text{waveform}} = \|\hat{x} - x\|_1.$$

Here, x denotes the clean target waveform and $\hat{x}$ denotes the restored waveform predicted by the model.

Model performance was evaluated using mean squared error (MSE), mean absolute error (MAE), signal-to-noise ratio (SNR), and scale-invariant signal-to-distortion ratio (SI-SDR). For MSE and MAE, lower values indicate better restoration. For SNR and SI-SDR, higher values indicate better restoration. In addition to these numerical metrics, waveform plots, spectrogram plots, and before and after audio files were generated to qualitatively compare the degraded, restored, and clean target signals.

4 Experimental Results and Discussion

The model was trained for 50 epochs. The training loss decreased from 0.11672 in epoch 1 to 0.08759 in epoch 50, indicating that the model learned to better map degraded waveform chunks toward the clean targets. The validation loss also improved overall, decreasing from 0.11134 in epoch 1 to a best validation loss of 0.09929 at epoch 31. After this point, the validation loss remained close to 0.10, suggesting that the model had mostly converged, however, since the training loss was slightly smaller than the validation loss, overfitting is suspected.

Table 1 summarizes the average test-set performance. The degraded column compares the original degraded audio to the clean target audio, while the restored column compares the WaveNet-restored audio to the clean target audio.

Table 1: Mean test-set audio restoration metrics. For MSE and MAE, lower is better. For SNR and SI-SDR, higher is better.

Metric	Degraded	Restored	Change	Interpretation
MSE	0.022680	0.006946	-0.015734	Error decreased
MAE	0.103067	0.049497	-0.053570	Error decreased
SNR	-5.177290	-0.140875	+5.036415	Signal quality improved
SI-SDR	-18.954794	-18.333614	+0.621180	Slight distortion improvement

The strongest improvements were observed in MSE and MAE. The MSE decreased from 0.022680 to 0.006946, which is about a 69% reduction in squared error. The MAE decreased from 0.103067 to 0.049497, which is about a 52% reduction in absolute waveform error. This suggests that the model learned a meaningful waveform level correction from the degraded recording toward the clean target.

The SNR also improved substantially, increasing by approximately 5.04 dB. This indicates that the restored waveform was closer to the clean signal than the original degraded recording. In practical terms, this means the model reduced some of the noise or distortion present in the degraded audio.

The SI-SDR improved only a little bit, increasing by approximately 0.62 dB. This smaller improvement suggests that although the model reduced sample-level error, it did not fully recover the perceptual structure of the clean speech. SI-SDR is stricter because it measures distortion after accounting for scale differences, so a small SI-SDR gain may indicate that the restored audio still contains artifacts or does not perfectly match the clean speech dynamics.

The waveform and spectrogram visualizations provide additional evidence for this interpretation. The restored waveform followed the clean target more closely than the degraded waveform in amplitude, but the spectrograms showed that the model did not perfectly reconstruct all frequency

components. This is expected because the model was trained on a limited subset of DAPS and used a simplified WaveNet style supervised setup rather than a full autoregressive WaveNet likelihood model.

Overall, the experiment shows that a WaveNet-style architecture can learn useful structure for dialogue restoration. However, the results also suggest that audio restoration is not only a waveform regression problem. A model can reduce MSE and MAE while still leaving perceptual artifacts, phase mismatch, or incomplete frequency reconstruction, meaning a hollow-sounding fix.

5 Conclusion

In this project, I attempted to implement a WaveNet-style neural network for restoring degraded dialogue using paired clean and degraded recordings from the DAPS dataset. The model used dilated one-dimensional convolutions, gated activations, residual connections, and skip connections to map degraded waveform chunks directly to clean waveform chunks.

The experimental results showed that the model was able to reduce waveform-level error. Compared with the degraded input audio, the restored audio had lower MSE, lower MAE, and higher SNR. This indicates that the model learned a meaningful transformation from consumer-device audio toward cleaner studio-quality speech. However, the smaller improvement in SI-SDR suggests that the restoration was not perfect and that the model still struggled with perceptual distortion and detailed frequency reconstruction.

These results support the idea that WaveNet-style models are useful for audio repair, but they also show why commercial audio isolation tools can sometimes produce hollow or robotic artifacts. Removing noise is not the same as fully reconstructing natural dialogue. Future work could improve this project by training on more degraded conditions, using longer temporal windows, adding perceptual audio losses, comparing against a U-Net or Demucs-style baseline, and evaluating the restored files with listening tests in addition to numerical metrics.

References

- [1] A. van den Oord, S. Dieleman, H. Zen, K. Simonyan, O. Vinyals, A. Graves, N. Kalchbrenner, A. Senior, and K. Kavukcuoglu, “WaveNet: A Generative Model for Raw Audio,” *arXiv preprint arXiv:1609.03499*, 2016.
- [2] G. J. Mysore, “Can We Automatically Transform Speech Recorded on Common Consumer Devices in Real-World Environments into Professional Production Quality Speech? – A Dataset, Insights, and Challenges,” *IEEE Signal Processing Letters*, vol. 22, no. 8, 2015. Dataset available as DAPS: Device and Produced Speech Dataset, Zenodo, DOI: 10.5281/zenodo.4660670.